

Some resources you may find useful besides the assigned text (considering many of the exercises are from these books) are Pinter's *A Book of Abstract Algebra* and Dummit and Foote's *Abstract Algebra*.

Definition of Groups

Exercise. Determine which of the following are groups. Verify the axioms or determine which do not hold.

1. Let $G = \{a, b\}$ and $\times : G \times G \rightarrow G$ be the binary operation given by $a^2 = a, b^2 = b, ab = a, ba = b$.
2. $(\mathbb{R}^+, \times)$.
3. The set of bijections from a set to itself (equipped with composition).
4. $(\mathbb{Z}/n\mathbb{Z}, +)$ for $n \in \mathbb{N}$ (note we need to check the operation is well-defined). What about $(\mathbb{Z}/n\mathbb{Z}, \times) \setminus \{0\}$?
5. The n th roots of unity $\{z \in \mathbb{C} : z^n = 1\}$ under addition. What about under multiplication?
6. $G = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ under addition. Check that G is not a group under multiplication, then consider $G \setminus \{0\}$ under multiplication.
7. The following subsets of $\mathbb{Q}$ under addition.
 - The set of rational numbers in lowest terms whose denominators are odd.
 - The set of rational numbers in lowest terms whose denominators are even.
 - The set of rational numbers with absolute value less than 1.
 - The set of rational numbers with absolute value greater than 1.
 - The set of rational numbers with denominators 1 or 2.
 - The set of rational numbers with denominators 1, 2, or 3.
8. The following subsets of $\mathbb{R}$ with the given operation.
 - All of $\mathbb{R}$ with $*$ as follows: Fix a constant k , and let $x * y = x + y + k$.
 - Nonzero numbers with $x * y = \frac{xy}{2}$.
 - $\mathbb{R} \setminus \{-1\}$ with $x * y = x + y + xy$.
 - $(-1, 1)$ with $x * y = \frac{x+y}{xy+1}$.

Exercise. Let S be a set. Consider the power set $P(S)$ endowed with the operation of symmetric difference, i.e. $A \Delta B = (A \setminus B) \cup (B \setminus A)$. Show this is a group. Find the identity element and describe inverses.

Exercise. Consider the set of binary strings of length n equipped with bitwise XOR. Show this is a group. Find the identity element and describe inverses.

Exercise. Let $(G, *)$ and $(H, \circ)$ be groups. Show that $G \times H$ is a group with the operation $(a, b) \cdot (c, d) = (a * c, b \circ d)$.

Basic Properties

Exercise. (On direct products) Let G, H be groups.

Show the order of $(a, b) \in G \times H$ is the LCM of the orders of a, b in G, H respectively (hint: Find commuting elements).

Show $G \times H$ is abelian if and only if G and H are.

Write the multiplication table for $\mathbb{Z}_2 \times \mathbb{Z}_3$. Does this group look familiar?

Exercise. (Self-inverses) Suppose G has even order. Show there exists x with $x^2 = e$ (hint: show first that for any group G , the number of elements not equal to their inverse is even).

Let $H = \{e, h_1, \dots, h_n\}$ be a finite abelian group. Let $a = h_1 \dots h_n$. Show that $a^2 = e$. Show that if there is no nontrivial element which is self-inverse, then $a = e$. If there is exactly one such element, show that a is equal to that element.

Show a group where every element is self-inverse must be abelian.

Exercise. (Groups of order 3 and 4) Show the following lemma.

Lemma. Let $a, b \in G$ be arbitrary.

- If $a^2 = a$, then $a = e$.
- If $ab = a$, then $b = e$. If $ab = b$, then $a = e$.

Use the lemma to show there is a unique group of order 3 (hint: make a multiplication table and show first that an element can only appear once in each row).

Let G be a group of order 4. Use the previous exercise to show there are only two groups of order 4: one with every element self inverse (this is known as the **Klein four-group**), and one with exactly one self inverse element. Do these groups look familiar?

Subgroups

Exercise. Let G be an abelian group. Show the following are subgroups of G . For 4-6, explain why the subsets are not subgroups if G is not abelian.

1. Elements which are self-inverse.
2. Elements x such that $x^n = e$ for some fixed natural number n .

3. Elements which have a square root.
4. Elements x such that $x^2 \in H$ for some fixed subgroup H .
5. More generally, elements x such that $x^n \in H$ for some integer n .
6. For two subgroups H, K of G , show the set $HK := \{hk : h \in H, k \in K\}$ is a subgroup of G .

Exercise. Let G be an arbitrary group.

1. Is the intersection of two subgroups a subgroup? What about the intersection of infinitely many?
2. Let H be a subgroup of G . If K is a subgroup of G contained in H , is K a subgroup of H ?
3. Show the **center**, the set of elements which commute with all elements of G , is a subgroup of G .
4. Assume G is finite. Show a subset S closed under multiplication is a subgroup (hint: For some $s \in S$, consider the distinct elements sg as g ranges over all elements in G).
5. Let H be a subgroup, and define $K = \{x : xax^{-1} \in H \text{ for all } a \in H\}$. Show K is a subgroup containing H .

Homomorphisms

Exercise. Show that the group of n th roots of unity is isomorphic to $\mathbb{Z}/n\mathbb{Z}$.

Show that $(\mathbb{R}^+, \times)$ and $\mathbb{R}$ are isomorphic.

Exercise. Determine which of the following, if any, are isomorphic to each other.

- E the group of even integers
- $\mathbb{Z}$
- $\mathbb{Q}$
- $\mathbb{R}$

Exercise. Let $\phi : G \rightarrow H$ be a homomorphism. Show $\phi(x^{-1}) = \phi(x)^{-1}$. Conclude the image of ϕ is a subgroup of H .

Exercise. An **automorphism** is an isomorphism from G to itself. Let G be a group. Show that the collection of automorphisms of G forms a group.

Are any of the following maps homomorphisms, and if so, are they automorphisms?

- $g \mapsto g^{-1}$

- $g \mapsto g^2$
- $g \mapsto x^{-1}gx$ for some fixed $x \in G$

Exercise. Show that the preimage of a subgroup is a subgroup. Furthermore, show the preimage of a normal subgroup is normal. Deduce that the kernel of a homomorphism is a normal subgroup.

Equivalence Relations

Exercise. Show the following are equivalence relations and describe the associated partitions.

1. $m \sim n$ if $|m| = |n|$ for $m, n \in \mathbb{Z}$
2. $r \sim s$ if $r - s \in \mathbb{Z}$ for $r, s \in \mathbb{Q}$
3. $a \sim b$ if $\text{floor}(a) = \text{floor}(b)$ for $a, b \in \mathbb{R}$
4. $m \sim n$ if $m - n$ is a multiple of 10 for $m, n \in \mathbb{Z}$
5. $a \sim b$ if $a - b \in \mathbb{Q}$ for $a, b \in \mathbb{R}$
6. $f \sim g$ if $f(0) = g(0)$ for $f, g : \mathbb{R} \rightarrow \mathbb{R}$
7. $f \sim g$ if $f(x) = g(x)$ for $x > c$ where c is some fixed real number and $f, g : \mathbb{R} \rightarrow \mathbb{R}$
8. Let C be a set and fix some subset D . For two subsets $A, B \in P(C)$, let $A \sim B$ if $A \cap D = B \cap D$.
9. $(a, b) \sim (c, d)$ if $a^2 + b^2 = c^2 + d^2$ for $(a, b), (c, d) \in \mathbb{R} \times \mathbb{R}$.
10. $a \sim b$ if $a/b \in \mathbb{Q}$ for $a, b \in \mathbb{R}$

Exercise. Let G be a group. Show the following are equivalence relations. Describe the equivalence class of the identity element.

1. Let H be a subgroup of G . Are the following two equivalence relations the same?
 - Let $a \sim b$ if $ab^{-1} \in H$.
 - Let $a \sim b$ if $a^{-1}b \in H$.
2. Let $a \sim b$ if $a = xbx^{-1}$ for some $x \in G$.
3. Let $a \sim b$ if $a^k = b^k$ for some integer k .
4. Let $a \sim b$ if ab^{-1} commutes with every element of G .
5. Let $a \sim b$ if ab^{-1} is a power of c for some fixed element $c \in G$.

Exercise. Let G be a group, H a subgroup, and $a, b \in G$. Prove the following.

1. $Ha = Hb$ iff $ab^{-1} \in H$.

2. $Ha = H$ iff $a \in H$.
3. If $aH = Ha$ and $bH = hB$, then $(ab)H = H(ab)$.
4. If $aH = Ha$, then $a^{-1}H = Ha^{-1}$.
5. If $(ab)H = (ac)H$, then $bH = cH$.
6. The number of left and right cosets of H agree.

Quotient groups

Exercise. (First Examples) $\mathbb{Z}$ and $\mathbb{Z}/n\mathbb{Z}$ are abelian, hence every subgroup is normal. Describe the quotients of $\mathbb{Z}$ and $\mathbb{Z}/n\mathbb{Z}$.

What is the quotient of $\mathbb{R}^2$ by the subgroup $(x, 0)$?

Find the normal subgroups of S_3 and the associated quotients.

Exercise. (More Examples) Show that $\mathrm{SL}_n(\mathbb{R})$ is a normal subgroup of $\mathrm{GL}_n(\mathbb{R})$ and describe the quotient.

The center of a group is normal. Find the center of $\mathrm{GL}_n(\mathbb{R})$ and describe the quotient.

Exercise. (Kernels and Quotients) Recall that kernels of homomorphisms are normal.

For each normal subgroup in the first exercise, find a homomorphism with kernel equal to the subgroup in question. Do the same for $\mathrm{SL}_n(\mathbb{R})$. What do you notice about these homomorphisms?

Exercise. Describe $\mathbb{Q}/\mathbb{Z}$ and $\mathbb{R}/\mathbb{Z}$.

To see a more concrete description, describe a homomorphism from $\mathbb{Q}, \mathbb{R}$ to $\mathbb{C}^\times$ with kernel $\mathbb{Z}$.

Exercise. Show the quotient of an abelian group is abelian. Give an example of a non-abelian group with abelian quotient.

Exercise. Show the center of a group is normal.

Show if the quotient of a group G by its center $Z(G)$ is cyclic, then the group is abelian (Hint: Pick a generator $aZ(G)$ of the quotient, then show every element of g is of the form ca^n , where c is in the center of G)

Group Actions and Products

Exercise. Compute the homomorphism associated to the action of K_4 on itself by left multiplication. Conclude there exists a subgroup of S_4 isomorphic to K_4 .

Exercise. Show the following are group actions.

- G acting on itself by left multiplication
- G acting on the cosets of a fixed subgroup H by left multiplication
- G acting on itself by conjugation

Exercise. Let H be a subgroup of G . Consider the action of G on the cosets of H by left multiplication. Compute the associated homomorphism in the case of D_8 acting on the cosets of a subgroup of order 2.

Show this action is transitive.

What is the stabilizer of H itself?

What is the kernel of the associated homomorphism $G \rightarrow S_n$?

Exercise. Recall a subgroup of index 2 is normal. Show the following more general statement.

Let H be a subgroup of G such that the index of H is the smallest prime dividing $|G|$. Show that H is normal.

Hint: Consider the action in the previous exercise. Determine the size of the kernel of the map $G \rightarrow S_p$ using the fact the index must divide $|G|$ and $|S_p|$. Consider the relationship between K and H to conclude.

Exercise. Let G, H be groups. Show the order of $(a, b) \in G \times H$ is the LCM of the orders of a, b in G, H respectively (Hint: Find commuting elements).

Exercise. Classify all groups of order 8 as follows. First, divide into the cases where G has an element of order 8, an element of order 4 (and no element of order 8), and an element of order 2 (and no elements of order 4 or 8).

The first and last are uniquely determined. For the remaining case, use what you know about products of groups to analyze the possible cases.

Exercise. Classify all groups of order $2p$ for a prime p .

(Try to) Classify all groups of order pq for primes p, q .

Sylow theorems

Exercise. (Some counterexamples) Show by counterexample that there it is not true that G contains a subgroup of order k for any $k \mid |G|$.

Hint: The smallest possible case works.

Show that Sylow's second theorem fails if we consider only p -subgroups rather than Sylow p -subgroups by considering the subgroups of order 4 in D_8 .

Exercise. (p -groups)

Recall that p -groups have nontrivial center.

Let H be a nontrivial normal subgroup of a p -group G . Show that a nontrivial normal subgroup intersects the center nontrivially (i.e. the intersection is not the trivial subgroup). Hint: Use the class equation, and use normality write $|H|$ as the sum of $|H \cap Z(G)|$ and conjugacy classes contained in H . Conclude using the fact that the sizes of the conjugacy classes divide $|G|$.

Furthermore, show if a normal subgroup has order p , then it is contained within the center.

(Optional) Let G be a p -group with order p^n . Show that G has a subgroup of order p^k for all $k \leq n$ by showing the same holds for any normal subgroup. Hint: Induct on n . Use the previous problem along with Cauchy's theorem to find a subgroup of order p and apply the correspondence theorem.

Exercise. (Sylow theorems and normal subgroups)

Show a group of order 147 has a normal 7-Sylow subgroup.

Show a group of order 495 has a normal subgroup. Hint: Consider the case where neither the 5-Sylow or 7-Sylow's are normal. How does a p -Sylow subgroup intersect its conjugates? Write $|G|$ as the sum of the orders of the p -Sylows minus the intersection and conclude.

Show a group of order 72 has a normal subgroup. Hint: Consider a 3-Sylow subgroup and let G act on the set of its conjugates (by conjugation) to get a map $G \rightarrow S_4$.

Basics of Rings

Exercise. Give examples of:

- A commutative ring with identity
- A commutative ring without identity
- A noncommutative ring with identity
- (Optional) A noncommutative ring without identity

Exercise. Determine the units in $\mathbb{Z}/8\mathbb{Z}$, $\mathbb{Z}/12\mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z}$.

Exercise. An element $x \in R$ is **nilpotent** if $x^k = 0$ for some positive integer k .

Show that if $n = a^k b$ then $\overline{ab}$ is nilpotent in $\mathbb{Z}/n\mathbb{Z}$.

Show that $\overline{a}$ is nilpotent in $\mathbb{Z}/n\mathbb{Z}$ iff every prime divisor of n is a prime divisor of a . Determine the nilpotent elements in $\mathbb{Z}/72\mathbb{Z}$.

Let R be the ring of functions from a nonempty set X to a field F . Show R contains no nonzero nilpotent elements.

Exercise. Let x be nilpotent.

Prove x is zero or a zero divisor.

Prove rx is nilpotent for all $r \in R$.

(Optional) Prove $1 + x$ is a unit and deduce the sum of a unit is a unit.

Exercise. Show that $S = \{(r, r) : r \in R\}$ is a subring of $R \times R$.

Exercise. Describe the homomorphic images of $\mathbb{Z}$.

Exercise. A ring has **characteristic** $k \in \mathbb{N}$ if $k1 := (1 + \dots + 1) = 0$. Show a ring has prime characteristic.

Let R have characteristic p . Show that $x \mapsto x^p$ is a ring homomorphism.

More on Rings

Exercise. Determine the maximal ideals of

- $\mathbb{Z}$
- $\mathbb{C}[x]$
- $\mathbb{R}[x]$
- $\mathbb{R} \times \mathbb{R}$
- $\mathbb{R}[x]/(x^2)$
- $\mathbb{R}[x]/(x^2 - 3x + 2)$
- $\mathbb{R}[x]/(x^2 + x + 1)$
- $\mathbb{Z}[x]$

Exercise. Find the units of $\mathbb{Z}/5\mathbb{Z}[x]$.

Find a nonconstant unit of $\mathbb{Z}/4\mathbb{Z}[x]$. What's the difference between these two cases?

Exercise. Are there fields k such that $k[x]/(x^2)$ and $k[x]/(x^2 - 1)$ are isomorphic?

Are $\mathbb{Z}/(6)$ and $\mathbb{Z}/(2) \times \mathbb{Z}/(3)$ isomorphic? What about $\mathbb{Z}/(8)$ and $\mathbb{Z}/(2) \times \mathbb{Z}/(4)$?